

# GeoLabeller

## Interface Control Document

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Software version v1.7.2

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This document specifies the external interfaces of GeoLabeller: the project file it reads and writes, the raster products it exports, the imagery it accepts, and the coordinate conventions all of these share. It is a description of the implemented behaviour, transcribed from the source.

# 1 Scope and Conventions

GeoLabeller is a desktop application for viewing georeferenced and non-georeferenced raster imagery and recording point annotations against it. This document specifies the interfaces through which other software exchanges data with it.

## 1.1 Interfaces covered

| Interface                      | Direction        | Section |
|--------------------------------|------------------|---------|
| Project file (.geoproj / JSON) | Read and written | 3       |
| Sub-image GeoTIFF export       | Written          | 4.1     |
| Optimized GeoTIFF export       | Written          | 4.2     |
| Source imagery                 | Read             | 5       |

The HDF5 training-dataset export is specified separately and is not repeated here.

This document ships with the application: an installed copy sits beside the executable and opens from Help > ICD, so the version on hand always describes the build in front of you.

## 1.2 Notation

- Field types are JSON types: number, string, boolean, array, object.
- Pixel coordinates are (column, row) with the origin at the top-left pixel of the source image; column increases right, row increases down. Fractional values are meaningful and are not rounded on write.
- Geographic coordinates are WGS84 decimal degrees (EPSG:4326), longitude and latitude reported separately and never packed.
- Distances in metres are geodesic (Haversine on WGS84), not planar distances in a projected CRS.

## 2 Coordinate Systems

Four coordinate systems appear across the interfaces. Only the first two are externally visible; the rest are described because they explain values that would otherwise look surprising.

| System                   | Units   | Where it appears                                                 |
|--------------------------|---------|------------------------------------------------------------------|
| WGS84 (EPSG:4326)        | degrees | Label lon/lat, waypoints, image corner coordinates               |
| Source pixel             | pixels  | Label pixel_x / pixel_y, relative to the unmodified source image |
| Web Mercator (EPSG:3857) | metres  | Internal display projection; not written to any file             |
| Scene                    | metres  | Internal canvas space; x = easting, y = -northing                |

### 2.1 Pixel coordinates are relative to the source image

Labels record the pixel they were placed on in the ORIGINAL image, at full resolution - not in the reprojected image shown on screen, and not at whatever pyramid level happened to be displayed. A label therefore addresses the same ground whatever the zoom was when it was placed, and a consumer can index the source raster directly.

### 2.2 Non-georeferenced imagery: the pixel zone

Imagery without a CRS cannot be placed geographically, so it is laid out in a reserved region of the internal scene beyond the valid extent of Web Mercator. This is an implementation detail with one external consequence: labels on such images carry pixel coordinates but no meaningful lon/lat, and those fields are written as 0.0.

| Constant             | Value      | Meaning                                              |
|----------------------|------------|------------------------------------------------------|
| PIXEL_ZONE_ORIGIN_X  | 25 000 000 | Scene X at which the non-georeferenced region begins |
| PIXEL_ZONE_SCALE     | 50         | Scene units per source pixel                         |
| PIXEL_ZONE_GROUP_GAP | 5 000      | Scene units between the columns of adjacent groups   |

### 2.3 Display resolution and memory

Imagery is displayed from pyramid overviews. The level chosen is the coarsest whose resolution is still no coarser than one screen pixel, subject to a cap on how much may be held at once. Images beyond that cap keep a coarse whole-image backdrop and read the detail for the visible area one tile at a time, so full resolution remains available at any zoom regardless of image size.

| Constant            | Value       | Meaning                                           |
|---------------------|-------------|---------------------------------------------------|
| MAX_LEVEL_PIXELS    | 150 000 000 | Largest whole-image level held in memory          |
| BACKDROP_MAX_PIXELS | 4 000 000   | Cap for the backdrop behind windowed detail tiles |
| TILE_SIZE           | 512         | Edge of a display tile, in destination pixels     |

## 3 Project File

A project is a single UTF-8 JSON document holding the class list, every image referenced, the labels on them, and any waypoints. Images are referenced by absolute path; their pixels are never copied into the project.

### 3.1 Top level

| Field             | Type            | Description                                                                                                |
|-------------------|-----------------|------------------------------------------------------------------------------------------------------------|
| version           | string          | Format version of this document, currently "3.7". Governs how label coordinates are interpreted - see 3.4. |
| classes           | array of string | Ordered class names. A label's class must appear here.                                                     |
| images            | array of object | One entry per referenced image; see 3.2.                                                                   |
| waypoints         | array of object | Named geographic bookmarks; see 3.5. Absent before version 3.3.                                            |
| _next_id          | number          | Next label id to allocate. Labels ids are unique per project.                                              |
| _next_waypoint_id | number          | Next waypoint id to allocate.                                                                              |

### 3.2 Image object

| Field                | Type            | Description                                                                                                                                                                                                                                                                                                                                    |
|----------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| path                 | string          | Absolute path to the image file. On load, a path that does not exist is first re-tried against the project file's own folder by matching progressively longer path tails, so a project that travels with its imagery opens unchanged on another machine; anything still missing can be relocated interactively (File > Locate Missing Images). |
| name                 | string          | File name without extension.                                                                                                                                                                                                                                                                                                                   |
| group                | string          | Group hierarchy, '/' separated; empty when ungrouped.                                                                                                                                                                                                                                                                                          |
| labels               | array of object | Labels on this image; see 3.3.                                                                                                                                                                                                                                                                                                                 |
| original_width       | number          | Source image width in pixels. Required to interpret label coordinates - see 3.4.                                                                                                                                                                                                                                                               |
| original_height      | number          | Source image height in pixels.                                                                                                                                                                                                                                                                                                                 |
| reader               | object          | Extension to reader name, e.g. {"tif": "default"}.                                                                                                                                                                                                                                                                                             |
| affine_coeffs        | array of number | Six coefficients of the source affine transform. Omitted when unknown.                                                                                                                                                                                                                                                                         |
| crs_epsg             | number          | EPSG code of the source CRS. Omitted for non-georeferenced imagery.                                                                                                                                                                                                                                                                            |
| corners_wgs84        | object          | Convenience only: top_left / top_right / bottom_right / bottom_left, each {lat, lon}. Written when the transform and CRS are known; derived, never read back.                                                                                                                                                                                  |
| geodesic_width_m     | number          | Convenience only: ground width along the top edge, metres.                                                                                                                                                                                                                                                                                     |
| geodesic_height_m    | number          | Convenience only: ground height along the left edge, metres.                                                                                                                                                                                                                                                                                   |
| hard_negative_source | boolean         | The image contains confusers but no true positives; the H5 export can opt in to sliding it whole into hard negatives. Present only when true; absent before version 3.5. Kept even while the image is not loaded, like labels.                                                                                                                 |

### 3.3 Label object

| Field      | Type   | Description                                                                                    |
|------------|--------|------------------------------------------------------------------------------------------------|
| id         | number | Sequential id, unique within the project.                                                      |
| unique_id  | string | UUID v4, unique to this label.                                                                 |
| class_name | string | Must appear in the top-level classes.                                                          |
| pixel_x    | number | Column in the source image. Stored as a FRACTION of original_width from version 2.1 - see 3.4. |

| Field              | Type   | Description                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pixel_y            | number | Row, stored the same way.                                                                                                                                                                                                                                                                                                                                                        |
| lon                | number | WGS84 longitude, or 0.0 for non-georeferenced imagery.                                                                                                                                                                                                                                                                                                                           |
| lat                | number | WGS84 latitude, or 0.0.                                                                                                                                                                                                                                                                                                                                                          |
| object_id          | string | UUID v4 shared by every label of the same real-world object. Labels are linked by sharing this value.                                                                                                                                                                                                                                                                            |
| length_m           | number | Measured object length in metres. Omitted when unmeasured.                                                                                                                                                                                                                                                                                                                       |
| width_m            | number | Measured width in metres. Omitted likewise.                                                                                                                                                                                                                                                                                                                                      |
| description        | string | Free text describing this labelled object. Belongs to this label alone and is never copied to labels sharing its object_id. Omitted when empty; absent before version 3.4.                                                                                                                                                                                                       |
| orientation_px_rad | number | Unit-circle principal angle of the object's drawn orientation in UN-WARPED source-pixel space: radians in $(-\pi, \pi]$ , x along the rows (+column), y up the columns. A vector from top-right to bottom-left is about $-3\pi/4$ (-135 degrees); rightward along a row is 0; up the image is $+\pi/2$ . Omitted when never drawn; absent before 3.7.                            |
| orientation_deg    | number | The same drawn vector as a TRUE-north heading: degrees clockwise from north in $[0, 360)$ , from a WGS84 geodesic azimuth of the endpoints through the image's transform - not grid north. Omitted for non-georeferenced imagery or when never drawn. Per label even when linked: two views of one object are two measurements.                                                  |
| group_id           | string | Human-readable name for the linked-object group - the readable companion to object_id. The inverse contract to description: it SHALL be identical across every label sharing an object_id. Updating it anywhere updates the whole group; a file that violates the invariant is repaired on load (first non-empty name in the group wins). Omitted when empty; absent before 3.6. |
| geodesic_distance  | number | Convenience only: metres from the image's left edge at the label's row. Derived, never read back.                                                                                                                                                                                                                                                                                |

### 3.4 Label coordinate storage (important)

From version 2.1, pixel\_x and pixel\_y are stored as FRACTIONS of the image dimensions, not as absolute pixels. A consumer must multiply by the image size to recover pixel coordinates:

```
pixel_x_absolute = pixel_x * original_width
pixel_y_absolute = pixel_y * original_height
```

The conversion applies only when the document version is 2.1 or later AND original\_width and original\_height are both greater than zero. If either is zero the values are absolute pixels and must be used unchanged. Both the writer and the reader apply this rule, so a file is self-consistent; it matters when reading a project with anything other than GeoLabeller itself.

### 3.5 Waypoint object

Waypoints are named geographic bookmarks belonging to the project rather than to any image. They are navigation aids and are never exported as training data.

| Field | Type   | Description                               |
|-------|--------|-------------------------------------------|
| id    | number | Sequential id, unique within the project. |
| name  | string | Display name, e.g. "WP 1".                |
| lat   | number | WGS84 latitude in degrees.                |
| lon   | number | WGS84 longitude in degrees.               |

### 3.6 Version history

| Version | Change                                                                                                              |
|---------|---------------------------------------------------------------------------------------------------------------------|
| 1.0     | Label-centric layout: a flat labels array, each entry naming its image. Still readable.                             |
| 2.0     | Image-centric layout: images each own their labels.                                                                 |
| 2.1     | Label pixel coordinates became fractions of the image dimensions (see 3.4).                                         |
| 3.2     | Adds derived convenience fields: corner coordinates and geodesic extents.                                           |
| 3.3     | Adds the top-level waypoints array and <code>_next_waypoint_id</code> . Files written before this simply have none. |
| 3.4     | Adds the optional per-label description string.                                                                     |
| 3.5     | Adds the optional per-image <code>hard_negative_source</code> flag.                                                 |
| 3.6     | Adds the shared per-group <code>group_id</code> string on labels.                                                   |
| 3.7     | Adds the optional <code>orientation_px_rad</code> and <code>orientation_deg</code> per label.                       |

Readers should compare the version as a string and treat anything at or above a threshold as supporting that feature. Older documents load unchanged; there is no migration step.

### 3.7 Example

A project with one image, one label and one waypoint. Note the fractional pixel coordinates: label 7 sits at absolute pixel (2048, 1024), being 0.5 x 4096 and 0.25 x 4096.

```
{
  "version": "3.4",
  "classes": ["vehicle", "building"],
  "images": [
    {
      "path": "D:/imagery/tile_014.tif",
      "name": "tile_014", "group": "survey/north",
      "original_width": 4096, "original_height": 4096,
      "reader": {"tif": "default"}, "crs_epsg": 32618,
      "labels": [
        {
          "id": 7, "class_name": "vehicle",
          "pixel_x": 0.5, "pixel_y": 0.25,
          "lon": -73.9832, "lat": 40.7536,
          "unique_id": "3b1e...", "object_id": "8f2c...",
          "length_m": 4.6, "width_m": 1.8,
          "description": "white pickup, tarpaulin over the bed"
        }
      ]
    }
  ],
  "waypoints": [{"id": 1, "name": "WP 1",
    "lat": 40.7536, "lon": -73.9832}],
  "_next_id": 8, "_next_waypoint_id": 2
}
```

## 4 Raster Exports

### 4.1 Sub-image export

Writes one GeoTIFF per label, cropped around it. The requested size is given in ground metres and converted to pixels using the true resolution at the label, so the crop covers the requested ground area whatever the CRS and whatever the pixel aspect.

| Property       | Behaviour                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------|
| Output path    | <output>/<class_name>/<object_id>_<label_id padded to 6>.tif                                                                 |
| Window         | Centred on the label's pixel, shifted (never cropped) to stay inside the image, so edge labels still yield a full-size crop. |
| Pixels         | Copied unmodified: all source bands, original data type, no rescaling.                                                       |
| Georeferencing | Source CRS preserved; transform adjusted to the window.                                                                      |
| Other metadata | nodata and per-band colour interpretation preserved.                                                                         |
| Compression    | LZW.                                                                                                                         |
| Skipped        | Labels on imagery with no CRS, labels outside the image, and windows that would be under half the requested size.            |

### 4.2 Optimized GeoTIFF export

Rewrites imagery as tiled, pyramided GeoTIFFs for fast display. Pixel values, band count, data type, CRS and transform are all preserved - only the storage layout changes.

| Property       | Behaviour                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output path    | Mirrors the group hierarchy under the output directory; keeps the source stem with a .tif extension.                                                                     |
| Layout         | Tiled, 512 x 512 blocks by default.                                                                                                                                      |
| Overviews      | Built to the requested decimation factors. Average resampling by default; paletted rasters always use nearest, so palette indices are never blended into invalid values. |
| Compression    | DEFLATE by default; may be set to another codec, none, or KEEP to retain the source's. A predictor is chosen to suit the data type.                                      |
| Large files    | BIGTIFF written when the result may exceed 4 GB.                                                                                                                         |
| Existing files | Skipped unless overwrite is requested. Output is written to a temporary file and moved into place, so an interrupted export leaves no partial .tif.                      |

## 5 Source Imagery

Imagery is read through GDAL, so any format GDAL can open may be loaded; file dialogs filter to .tif and .tiff by default, and directory import scans for those extensions.

### 5.1 Georeferenced and non-georeferenced imagery

An image with a CRS is reprojected for display and its labels carry both pixel and WGS84 coordinates. An image without one is shown at its own pixel scale in the region described in 2.2, and its labels carry pixel coordinates only, with lon and lat written as 0.0.

### 5.2 Recommendations

- Supply tiled GeoTIFFs with overviews. Without a pyramid the whole image must be decoded before anything can be shown; with one, a coarse level appears immediately and sharpens. Section 4.2 produces suitable files.
- 8-bit imagery is displayed exactly as stored. Other data types are linearly stretched for display between the 2nd and 98th percentile of a sampled read; the underlying values are never modified.
- Very large mosaics are supported: detail for the visible area is read a tile at a time, so display cost follows the viewport rather than the size of the image.

## 6 Operation

This section is included because the way a label is produced decides which fields it carries. A consumer reading a project file will meet labels that share an `object_id`, labels with measurements and labels whose lon/lat are zero, and 6.3 explains where each comes from.

### 6.1 Interaction modes

The canvas is always in exactly one mode. Modes differ in what a left click does and how the view is navigated; none of them change how a label is stored.

| Mode       | Left click          | Purpose                                                                                                             |
|------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Pan        | Drags the view      | Navigating. Right click also offers waypoint and layer actions.                                                     |
| Label      | Places a label      | Ordinary labelling of whatever is on screen.                                                                        |
| Cycle      | Places a label      | Steps through a group one image at a time; Space advances.                                                          |
| View Cycle | Places a label      | As Cycle, but over the images currently in view.                                                                    |
| Waterfall  | Places a label      | Stacks a bottom-level group vertically as raw pixels and glides through it; every label in the group stays visible. |
| Ruler      | Drags a measurement | Reading ground distances. Records nothing in the project.                                                           |

Two overlays work on top of whichever labelling mode is active, rather than replacing it:

| Overlay               | Effect                                                                                                                                                                                                                                                                                                         |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chain Link            | Clicking labels one after another links them into a single object; each click writes the shared <code>object_id</code> immediately.                                                                                                                                                                            |
| Measure               | Draws a length and then a width line on a label, writing <code>length_m</code> and <code>width_m</code> . Georeferenced imagery only, since metres are undefined without a CRS.                                                                                                                                |
| Description           | Right click a label to write a free-text note on it, shown as the marker's tooltip. Stays with that label when it is linked.                                                                                                                                                                                   |
| Locate missing images | File menu. Rebinds a shared project's images to their paths on this machine: search a folder (filename matches verified against each image's recorded size and CRS) or locate one file and apply its old-versus-new path prefix to the rest. Applied paths persist when the project is saved.                  |
| Orient                | Labels > Orientation Editor shows a class's snippets in a grid; dragging tail-to-nose across one stores <code>orientation_px_rad</code> and, for georeferenced imagery, <code>orientation_deg</code> . Snippets are raw source pixels with the export's framing, so the drawn vector is a source-pixel vector. |
| Flag hard negatives   | Right click an image to mark it as a hard-negative source: it holds confusers but no true positives. Flagged images collect in the Hard Negatives panel, and the H5 export can include them as <code>gt=false</code> hard negatives even under a labels-only scope.                                            |

### 6.2 How a label's fields are produced

Every field in 3.3 is written for every label; these are the ones whose value depends on how or where the label was made.

| Field                                        | Produced by                                            | Consumer note                                                                                                                                               |
|----------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>pixel_x</code> , <code>pixel_y</code>  | Any left click in a labelling mode                     | Always the source image's own pixels at full resolution, whatever the zoom or pyramid level on screen. Stored as fractions - see 3.4.                       |
| <code>lon</code> , <code>lat</code>          | Derived from the clicked pixel through the image's CRS | Zero when the image has no CRS. Check <code>crs_epsg</code> on the image before trusting these.                                                             |
| <code>object_id</code>                       | Shared when labels are linked                          | Unique per label until then. Labels of one real object are found by grouping on this value, not by proximity.                                               |
| <code>length_m</code> , <code>width_m</code> | Measure overlay                                        | Absent when never measured. Geodesic metres, not planar. Copied to every label sharing the <code>object_id</code> when the linked-measurement option is on. |

| Field       | Produced by                                | Consumer note                                                                                                                                                                                                                                            |
|-------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| description | Right click a label, Description...        | Free text, absent when empty. Per label even when labels are linked: the same object in two images is described separately.                                                                                                                              |
| group_id    | Right click a label, Group ID...           | One value for the whole linked group, always: setting it on any member applies it to every label sharing the object_id. A label that is unlinked loses it; a label joining a named group adopts the name (the first label's name wins a merge conflict). |
| class_name  | The class selected when the click was made | Always one of the top-level classes.                                                                                                                                                                                                                     |

### 6.3 Keyboard and mouse reference

Generated from the application's own binding table, so it matches the reference shown by F1 exactly.

| File Operations |                              |
|-----------------|------------------------------|
| Ctrl+N          | New Project                  |
| Ctrl+Shift+P    | Open Project                 |
| Ctrl+S          | Save Project                 |
| Ctrl+Shift+S    | Save Project As              |
| Ctrl+O          | Add Image (GeoTIFF + custom) |
| Ctrl+Shift+O    | Add Directory                |
| Ctrl+Q          | Exit                         |

| Navigation   |                             |
|--------------|-----------------------------|
| Mouse Wheel  | Zoom in/out                 |
| Click + Drag | Pan (in Pan mode)           |
| Right-click  | Context menu                |
| Ctrl+G       | Go to a latitude/longitude  |
| Ctrl+Shift+W | Add waypoint by coordinates |

| Mode Switching |                                         |
|----------------|-----------------------------------------|
| P              | Pan mode                                |
| L              | Label mode                              |
| C              | Cycle mode (group-based)                |
| V              | View Cycle (layers in view)             |
| W              | Waterfall mode (stack group vertically) |
| R              | Ruler mode                              |

| Labeling (Label & Cycle modes) |                                       |
|--------------------------------|---------------------------------------|
| Left-click                     | Place label                           |
| Right-click label              | Label options (remove, link, measure) |
| Right-click + drag             | Pan the view                          |
| Ctrl+Left-click label          | Label options (Cycle modes)           |
| 1-9                            | Quick-switch to class 1-9             |
| Escape                         | Cancel link mode                      |

| Chain Linking    |                                       |
|------------------|---------------------------------------|
| K                | Toggle chain-link (any labeling mode) |
| Left-click label | Add to chain (first click anchors)    |
| N                | Finish this chain, start a new one    |
| Escape           | Exit chain-link (links are kept)      |

| Cycle Modes (Cycle / View Cycle) |                                       |
|----------------------------------|---------------------------------------|
| Space                            | Next layer (unchecks current)         |
| Ctrl+Space                       | Go back to previous layer             |
| Left-click                       | Place label                           |
| Right-click label                | Label options (remove, link, measure) |
| Right-click + drag               | Pan around                            |
| Mouse wheel                      | Zoom in/out                           |

| Waterfall Mode     |                                         |
|--------------------|-----------------------------------------|
| Hold Space         | Glide up through the image stack        |
| Hold Ctrl+Space    | Glide down (starts at the bottom)       |
| Left-click         | Place label (group labels stay visible) |
| Right-click label  | Label options (remove, link, measure)   |
| Right-click + drag | Pan within the stack                    |
| Mouse wheel        | Zoom in/out                             |

| Measuring    |                                            |
|--------------|--------------------------------------------|
| Shift + Drag | Measure distance, without changing mode    |
| M            | Measure label under cursor (length, width) |
| Escape       | Clear the measurement or the go-to marker  |

| Waypoints             |                                    |
|-----------------------|------------------------------------|
| Right-click map       | Add waypoint here (Pan mode, geo)  |
| Ctrl+Shift+W          | Add waypoint by coordinate         |
| Right-click marker    | Go to / Rename / Remove            |
| Double-click in panel | Fly to that waypoint               |
| Show on map           | Hide or show every waypoint marker |

| Layer Panel       |                                      |
|-------------------|--------------------------------------|
| Checkbox          | Toggle layer/group visibility        |
| Right-click group | Select/Unselect, Expand/Collapse all |
| Right-click layer | Zoom to layer, Remove                |
| Drag & Drop       | Reorder layers/groups                |

| Labeled Images Panel |                                 |
|----------------------|---------------------------------|
| Checkbox             | Toggle image visibility (syncd) |
| Right-click label    | Zoom to label or layer          |
| Right-click group    | Select/Unselect all in group    |

| Help |                |
|------|----------------|
| F1   | Show this help |